

Systematic Evaluation of Clustering Methods for scRNA-seq Data

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Abstract

Single-cell RNA sequencing (scRNA-seq) data pose significant challenges due to high dimensionality, sparsity, and noise, making accurate cell clustering difficult. This project systematically evaluates multiple state-of-the-art representation learning and clustering approaches, such as IGCLAPS, scCRT, scGCL, and UIPBC, to understand how contrastive learning, transformer-based encoding, graph learning, and classical pipelines affect clustering quality. Performance is assessed using standard metrics such as ARI, NMI, and Purity across publicly available scRNA-seq datasets.

Objective

- To implement and understand multiple scRNA-seq clustering frameworks based on different learning paradigms.
- To evaluate clustering performance using external evaluation measures.

Background Study

- IGCLAPS:** A graph-based contrastive learning framework that learns cell representations by contrasting positive and negative graph views.
- scCRT:** A transformer-based model that captures global gene dependencies using self-attention mechanisms.
- scGCL:** A graph contrastive learning approach that leverages cell-cell relationships for robust embedding learning.
- UIPBC:** A pipeline-based clustering method relying on dimensionality reduction (PCA) followed by clustering and evaluation.

Methodology

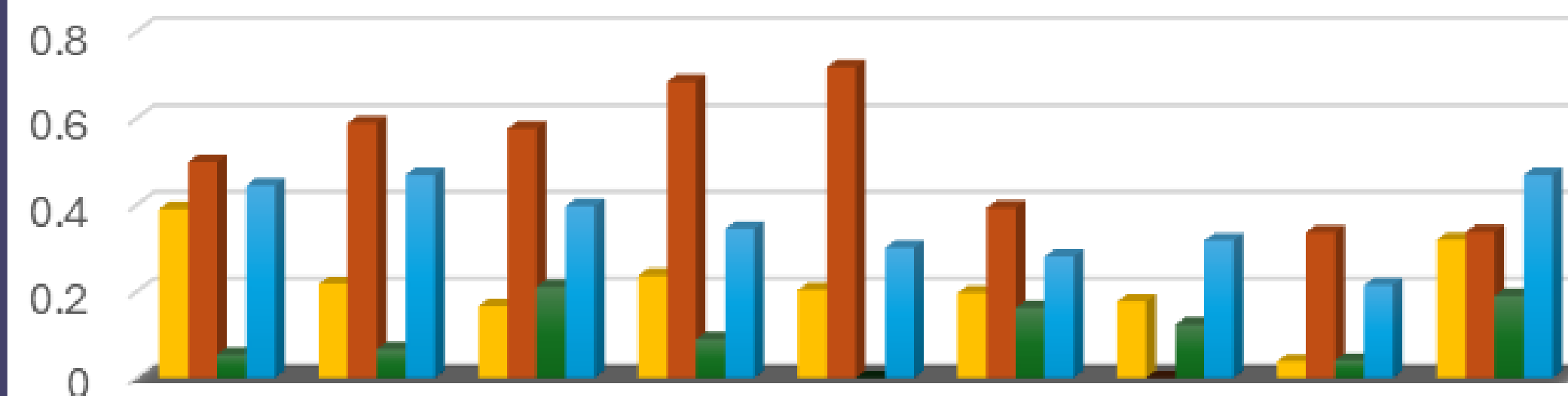
- Implemented **IGCLAPS**, **scCRT**, **scGCL**, and **UIPBC** clustering methods.
- Preprocessed scRNA-seq datasets (normalization, filtering, feature selection) to deal with noise, batch effects, data distribution, and transcript depths.
- Evaluated clustering results using **Adjusted Rand Index (ARI)**, **Normalized Mutual Information (NMI)**, and **Purity**, and identified best method.

Datasets

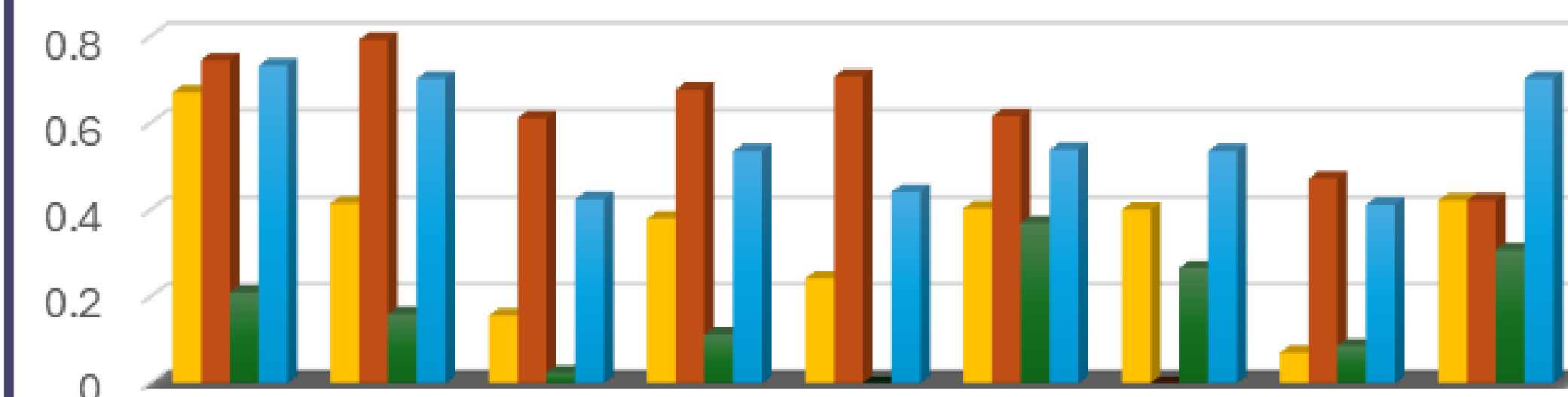
Name	Cells	Genes	Clusters
GSE45719	291	45686	14
GSE48968	1378	45686	5
GSE52529	288	65218	3
GSE63818	328	65218	5
GSE64016	460	65218	4
GSE71585	749	45686	18
GSE75140	729	35238	13
GSE81076	1152	65218	12
GSE57872	864	65218	9

Results

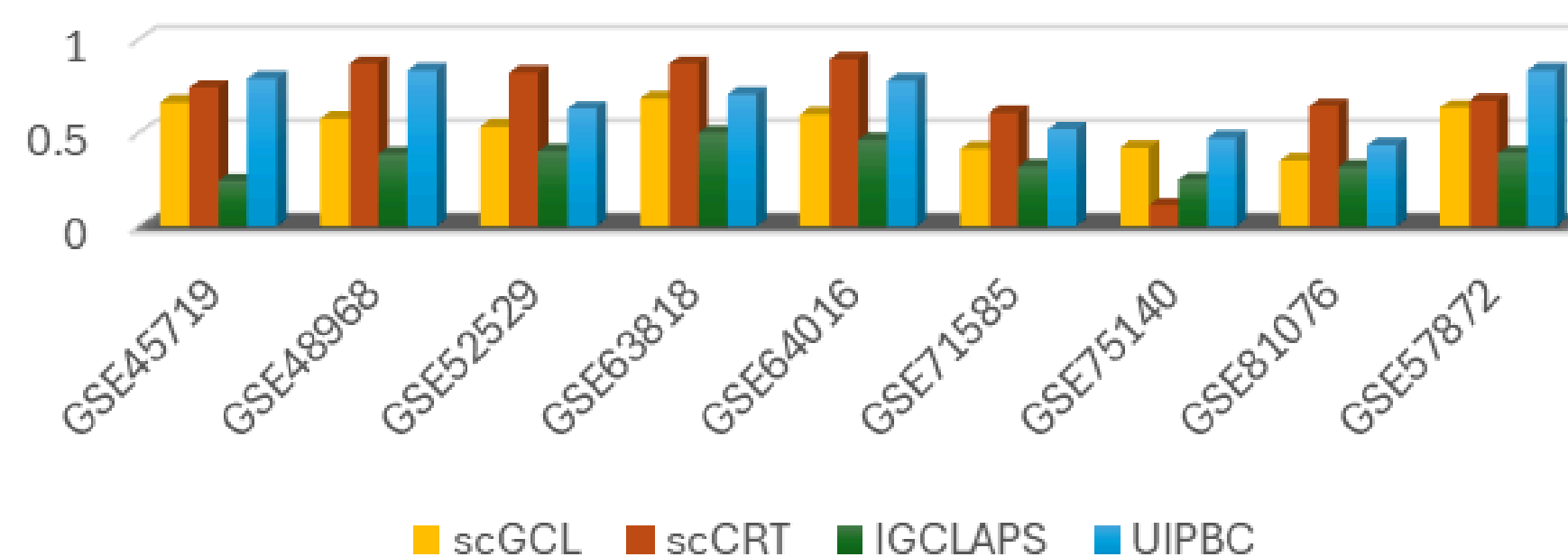
Performance comparison in terms of ARI



Performance comparison in terms of NMI

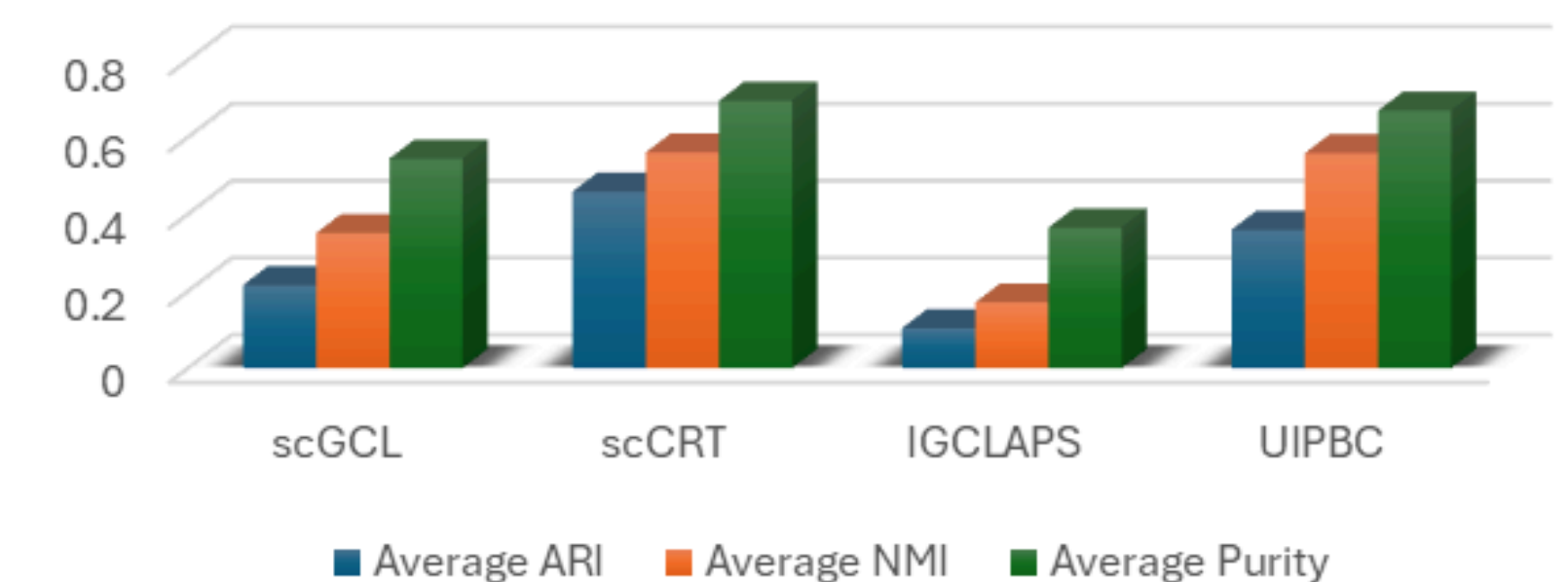


Performance comparison in terms of Purity



- Top Performer:** scCRT performed best on ~7/9 datasets, achieving peak scores of ARI ≈ 0.73 , NMI ≈ 0.80 , and Purity ≈ 0.90 .
- Deep Learning Baseline:** scCRT consistently outperformed IGCLAPS and scGCL, showing ~15–30% higher ARI/NMI on most datasets.
- Stability:** scCRT showed lower variance across datasets, while IGCLAPS and scGCL were more sensitive to sparsity and dataset size.
- Noise Reduction:** Higher Purity (≈ 0.80 – 0.90) indicates stronger noise suppression; UIPBC remained a stable but weaker baseline.

Performance comparison in terms of Average



Key Insights

- Transformer-based models provide stronger noise robustness than local graph-based approaches.
- Representation quality dominates clustering performance, often more than the choice of the clustering algorithm itself.
- Stability across datasets is a more reliable indicator of real-world usability than peak performance on individual benchmarks.

Conclusion & Future Scope

- Comparative evaluation across IGCLAPS, scCRT, scGCL, and UIPBC provides practical insights into method selection for different data characteristics.
- Future Scope:** Integrating hybrid transformer-graph architectures and validating results with biological annotations and downstream analyses.

References

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- [2] Xiong, Z. et al. (2023). scGCL: an imputation method for scRNA-seq data based on graph contrastive learning. Bioinformatics, 39(3).
- [3] Zheng, W. et al. (2025). IGCLAPS: an interpretable graph contrastive learning method with adaptive positive sampling for scRNA-seq data analysis. Bioinformatics, 41(8).
- [4] Shi, Y. et al. (2024). scCRT: a contrastive-based dimensionality reduction model for scRNA-seq trajectory inference. Briefings in Bioinformatics, 25(3).